ParkChain

A Smart-Contract-Based Urban Parking and EV Charging Network

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Introduction to ParkChain

Urban parking and EV charging services are often fragmented across multiple providers. Users must manage different reservation systems and payment methods, while operators require trustworthy accounting and revenue management.

The goal of ParkChain is to provide a decentralized platform for parking reservation and EV charging management using Ethereum smart contracts and blockchain-based credits.

Stakeholders

Member

Purchases memberships, receives ParkCredits, reserves parking slots, checks in, checks out, and consumes charging services.

Operator

Provides parking and charging infrastructure, configures pricing models, and receives revenue through the platform treasury.

Administrator

Registers operators, manages membership plans, configures system-wide parameters, and supervises platform governance.

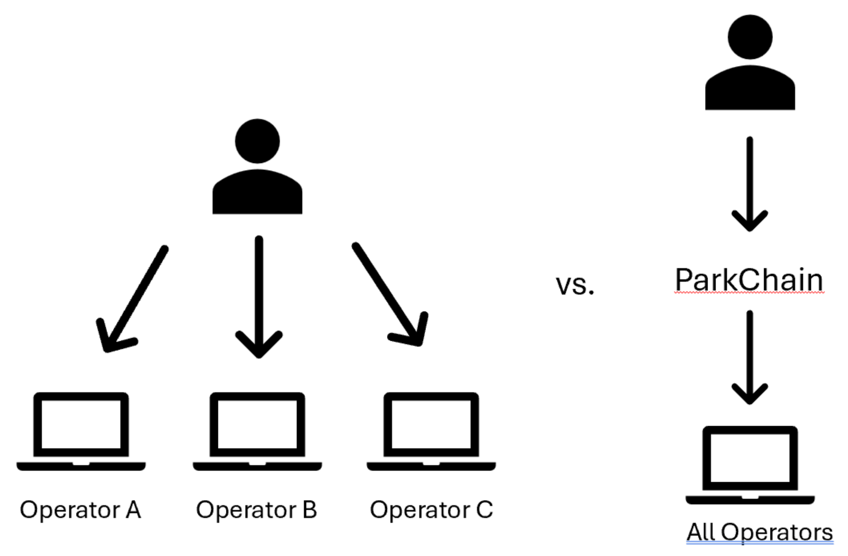


Figure 1. Enter Caption

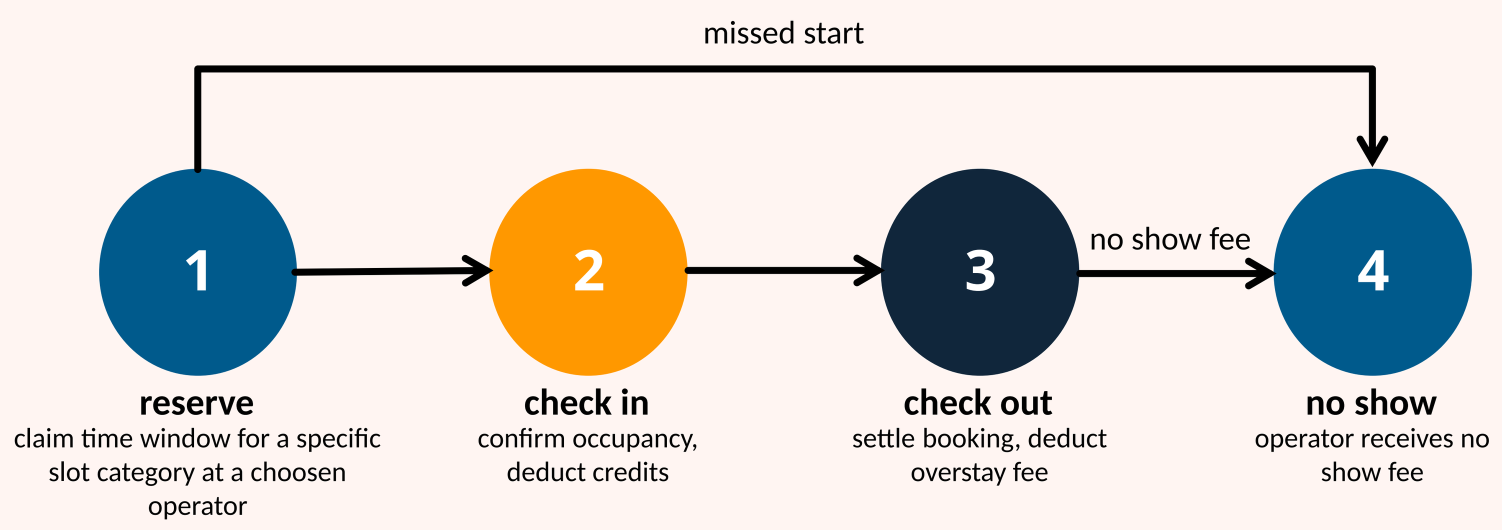
System Concept

ParkChain uses Ethereum smart contracts and ERC-1155-compatible ParkCredits as the platform's payment and accounting mechanism. Members purchase monthly memberships that mint ParkCredits to their wallet. These credits are then used to reserve parking slots or EV charging bays across all registered operators. The platform supports multiple membership tiers with different:

- credit allowances,
- monthly occupancy limits,
- and usage restrictions per operator and slot category

Reservation lifecycle

Parking follows a four-phase lifecycle:



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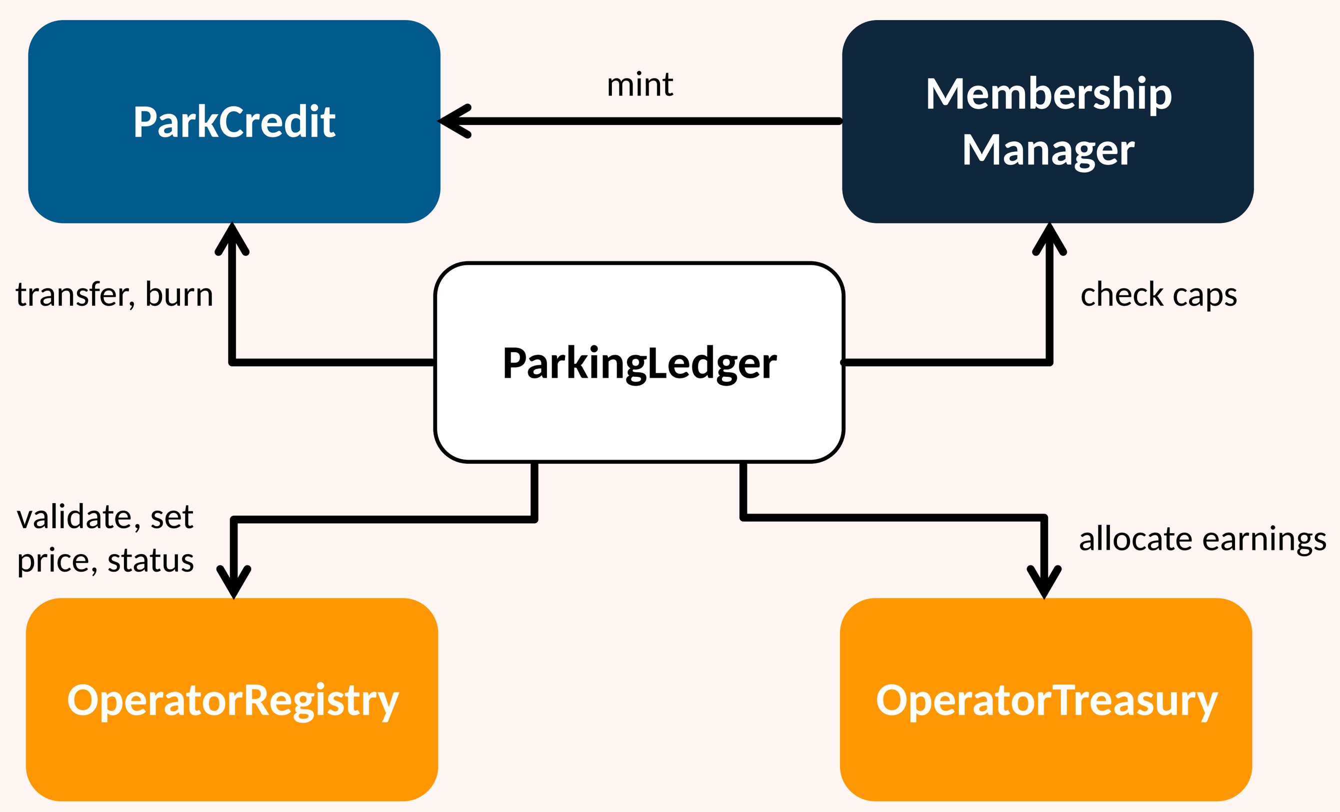
Contract Modules



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Challenges and solutions

Challenges

- Integration of independently developed components
- Merge conflicts during contract development
- Coordinating interfaces between smart contracts

Solutions

- Regular synchronization meetings
- Clear contract responsibilities
- Incremental integration and testing

Current Progress

- BPMN modeling
- Stakeholder and smart contracts analysis
- Implement tokens, operator registry and treasury
- Implement tests and running on local devices
- Initial frontend structure

Next steps

- Complete ParkingLedger
- Complete MembershipManager
- Implement reservation logic
- Integration tests between contracts
- Frontend integration

References

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